

# Overture 3

## Digital Filter Theory & Mathematical Formulations

Synthesis Equations, Prototype Approximations, and Bilinear Transformations

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### Abstract

This document provides the formal mathematical foundations underlying the DSP synthesis engine in **Overture 3**. It outlines continuous-time analog prototype transfer functions, closed-form pole-zero calculations, conformal Bilinear Transform (BLT) mappings with frequency pre-warping, and discrete-time cascaded Biquad Second-Order Sections (SOS) implemented in zero-allocation modern C++20.

## 1 Introduction and Discrete Biquad Cascade Architecture

In digital audio and precision signal processing, higher-order IIR filters are synthesized as a cascade of Second-Order Sections (SOS) to avoid numerical instability and severe coefficient quantization errors. The composite transfer function  $H(z)$  is factored into  $K = \lceil N/2 \rceil$  biquadratic sections:

$$H(z) = g \cdot \prod_{k=1}^K \frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \quad (1)$$

where  $g$  is the overall system gain. In Overture 3, each section is realized using the **Direct Form II Transposed** structure, described by the state-space difference equations:

$$y_k[n] = b_{0k}x_k[n] + w_{1k}[n-1] \quad (2)$$

$$w_{1k}[n] = b_{1k}x_k[n] - a_{1k}y_k[n] + w_{2k}[n-1] \quad (3)$$

$$w_{2k}[n] = b_{2k}x_k[n] - a_{2k}y_k[n] \quad (4)$$

The filter is Bounded-Input Bounded-Output (BIBO) stable if and only if all poles  $p_{ik}$  reside strictly inside the complex unit circle:

$$|p_{ik}| < 1 \quad \forall i \in \{1, 2\}, k \in \{1, \dots, K\} \quad (5)$$

## 2 Analog Prototype Approximations

### 2.1 Butterworth Approximation (Maximally Flat)

The Butterworth filter exhibits a maximally flat magnitude response in the passband with no ripple. The squared magnitude response of an  $N$ -th order analog prototype with cutoff  $\Omega_c$  is:

$$|H(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad (6)$$

The  $2N$  poles of  $H(s)H(-s)$  lie symmetrically on a circle of radius  $\Omega_c$ . The  $N$  stable left-half plane poles are given in closed form by:

$$s_k = \Omega_c \exp\left(j \left[ \frac{(2k + N - 1)\pi}{2N} \right]\right), \quad k = 1, 2, \dots, N \quad (7)$$

## 2.2 Chebyshev Type I (Passband Equiripple)

Chebyshev Type I filters provide a steeper transition roll-off than Butterworth filters of the same order by permitting equiripple variation  $\epsilon$  in the passband:

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2\left(\frac{\Omega}{\Omega_c}\right)}, \quad \epsilon = \sqrt{10^{\delta_{\text{pass}}/10} - 1} \quad (8)$$

where  $T_N(\omega)$  denotes the  $N$ -th order Chebyshev polynomial of the first kind:

$$T_N(\omega) = \begin{cases} \cos(N \arccos \omega), & |\omega| \leq 1 \\ \cosh(N \operatorname{arcosh} \omega), & |\omega| > 1 \end{cases} \quad (9)$$

The prototype poles lie on an ellipse with semi-minor axis  $a = \Omega_c \sinh(\gamma)$  and semi-major axis  $b = \Omega_c \cosh(\gamma)$ , where  $\gamma = \frac{1}{N} \operatorname{arsinh}(1/\epsilon)$ :

$$s_k = -\Omega_c \sinh(\gamma) \sin\left(\frac{(2k-1)\pi}{2N}\right) + j \Omega_c \cosh(\gamma) \cos\left(\frac{(2k-1)\pi}{2N}\right), \quad k = 1, \dots, N \quad (10)$$

## 2.3 Chebyshev Type II (Inverse Chebyshev)

Chebyshev Type II filters feature a maximally flat passband and equiripple stopband attenuation  $\delta_{\text{stop}}$  (in dB), with transmission zeros on the imaginary axis:

$$|H(j\Omega)|^2 = \frac{1}{1 + \frac{1}{\epsilon_s^2 T_N^2(\Omega_s/\Omega)}}, \quad \epsilon_s = \frac{1}{\sqrt{10^{\delta_{\text{stop}}/10} - 1}} \quad (11)$$

Poles are the reciprocal inversion of Type I poles, and the finite transmission zeros occur at:

$$z_m = j \frac{\Omega_s}{\cos\left(\frac{(2m-1)\pi}{2N}\right)}, \quad m = 1, \dots, \lfloor N/2 \rfloor \quad (12)$$

## 2.4 Elliptic / Cauer Approximation

Elliptic filters achieve the steepest transition roll-off for a given order  $N$  by allowing equiripple behavior in both passband and stopband:

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 R_N^2(\xi, \Omega/\Omega_c)} \quad (13)$$

where  $R_N(\xi, \omega)$  is the Chebyshev rational function and  $\xi \geq 1$  is the selectivity factor. Zeros lie purely on the  $j\Omega$ -axis, providing deep attenuation nulls.

## 2.5 Bessel-Thomson (Linear Phase / Maximally Flat Group Delay)

Designed to preserve audio transient waveforms and impulse shapes without overshoot or ringing, the Bessel transfer function is:

$$H(s) = \frac{\theta_N(0)}{\theta_N(s/\omega_0)} \quad (14)$$

where  $\theta_N(s)$  is the reverse Bessel polynomial defined via the recurrence:

$$\theta_0(s) = 1, \quad \theta_1(s) = s + 1, \quad \theta_n(s) = (2n-1)\theta_{n-1}(s) + s^2\theta_{n-2}(s) \quad (15)$$

The group delay  $\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega}$  is maximally flat at DC ( $\omega = 0$ ).

### 3 Bilinear Transform and Frequency Pre-Warping

The continuous  $s$ -domain prototype is mapped to the discrete  $z$ -domain via the Bilinear Transform:

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}}, \quad T_s = \frac{1}{F_s} \quad (16)$$

To eliminate non-linear frequency compression near the Nyquist frequency  $F_s/2$ , critical analog frequencies  $\Omega$  are pre-warped from the target digital frequencies  $\omega = 2\pi F_c/F_s$ :

$$\Omega = \frac{2}{T_s} \tan\left(\frac{\omega T_s}{2}\right) = 2F_s \tan\left(\frac{\pi F_c}{F_s}\right) \quad (17)$$

For bandpass and bandstop filters, both lower ( $F_{c1}$ ) and upper ( $F_{c2}$ ) cutoff frequencies are independently pre-warped to preserve the exact center frequency  $\omega_0 = \sqrt{\omega_1\omega_2}$  and bandwidth  $B = \omega_2 - \omega_1$ .

### 4 References & Canonical Literature

- [Butterworth Filter Theory — Wikipedia](#)
- [Chebyshev Filters \(Type I & II\) — Wikipedia](#)
- [Elliptic \(Cauer\) Rational Filter Synthesis — Wikipedia](#)
- [Bessel-Thomson Linear Phase Filters — Wikipedia](#)
- [Bilinear Transform with Pre-Warping — Wikipedia](#)
- [Digital Biquad Second-Order Sections \(SOS\) — Wikipedia](#)
- [Overtune 3 GitHub Repository & DSP Source Code](#)